

Nmap Network Reconnaissance Report

Target Vulnerability Discovery & Port Enumeration

Task 2: Cybersecurity
Internship
Target: 10.0.2.4

Scan Execution Metadata

Command Executed	nmap -sS -sV -O -oN nmap_scan.txt 10.0.2.4
Initiated / Completed	Fri Aug 28 19:50:46 2026 – 19:51:52 2026 (65.39s)
Target Status	Host is up (0.0023s latency, 1 hop)
Network Interface	Oracle VirtualBox Virtual NIC (MAC: 08:00:27:E0:77:F0)

Target Operating System

Device Type: General Purpose
Running OS: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS Details: Linux 2.6.9 - 2.6.33 (Ubuntu 8.04 LTS)

Host Identifiers & Scope

Hostnames: metasploitable.localdomain,
irc.Metasploitable.LAN
Total Ports Scanned: 1,000 TCP ports
Closed Ports: 977 (filtered/reset)
Open Ports Discovered: 23

Port Scan Summary Statistics

Out of 1,000 probed TCP ports, 977 were found closed or filtered, leaving an exceptionally broad surface area of **23 open ports** running legacy daemons and network services.

Discovered Services (Ports 21 – 1099)

Comprehensive enumeration of active TCP ports, protocols, and identified software versions.

Port	State	Service	Version / Fingerprint Details
21/tcp	OPEN	ftp	vsftpd 2.3.4
22/tcp	OPEN	ssh	OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp	OPEN	telnet	Linux telnetd (Cleartext protocol)
25/tcp	OPEN	smtp	Postfix smtpd
53/tcp	OPEN	domain	ISC BIND 9.4.2
80/tcp	OPEN	http	Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp	OPEN	rpcbind	2 (RPC #100000)
139/tcp	OPEN	netbios-ssn	Samba smbd 3.X - 4.X (WORKGROUP)
445/tcp	OPEN	netbios-ssn	Samba smbd 3.X - 4.X (WORKGROUP)
512/tcp	OPEN	exec?	Unverified remote execution service
513/tcp	OPEN	login	rlogin service
514/tcp	OPEN	tcpwrapped	TCP wrapper protected service daemon
1099/tcp	OPEN	java-rmi	GNU Classpath grmiregistry

Discovered Services (Ports 1524 – 8180)

Port	State	Service	Version / Fingerprint Details
1524/tcp	OPEN	bindshell	Metasploitable root shell (Critical Backdoor)
2049/tcp	OPEN	nfs	2-4 (RPC #100003)
2121/tcp	OPEN	ftp	ProFTPD 1.3.1
3306/tcp	OPEN	mysql	MySQL 5.0.51a-3ubuntu5
5432/tcp	OPEN	postgresql	PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp	OPEN	vnc	VNC (protocol 3.3)
6000/tcp	OPEN	X11	X11 display server (access denied)
6667/tcp	OPEN	irc	UnrealIRCd
8009/tcp	OPEN	ajp13	Apache Jserv (Protocol v1.3)
8180/tcp	OPEN	http	Apache Tomcat/Coyote JSP engine 1.1

Key Security Observations

- **Expanded Attack Surface:** Exposure of 23 concurrent network services provides an overly broad attack vector profile.
- **Plaintext Management Protocols:** Active Telnet (23), rlogin (513), and multiple FTP implementations transmit passwords in cleartext.
- **Pre-configured Backdoor Shells:** Port 1524 exposes an unauthenticated root bind shell directly on the network interface.
- **Exposed Back-End Databases:** Unrestricted network access to MySQL (3306) and PostgreSQL (5432) increases susceptibility to brute-force attacks.